

— THE COMPLETE TOUR

JanVayu, end to end

Every section of the platform — the live data, the health toll, the money, the accountability, and the tools — one slide at a time.

— CHAPTER 01

Why it exists

01

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India breathes the world's most dangerous air

- **1.72 million** deaths a year from ambient PM2.5 (Lancet Countdown 2025)
- **3.5 years** of life expectancy lost on average (AQLI 2025)
- **6th** most-polluted country; Loni the most-polluted city on Earth (IQAir 2025)
- Roughly a quarter of the global PM2.5 death burden — the largest of any nation

The scale is why JanVayu exists: to turn scattered official data into a public record anyone can act on.

A public record, not a campaign

- **Non-partisan** — we track promises and money, not parties
- **Citizen-led and open** — anyone can read, reuse or improve it
- **Every number cites its source**, and is re-checked against it every week
- Free, forever, no ads

The case for clean air is strong enough that it never needs an exaggerated number.

One clear picture, from many sources

- **CPCB / CAAQMS** — India's official monitoring network
- **WAQI, IQAir, OpenAQ** — global aggregators and community sensors
- **CREA, WHO, Lancet, NASA** — research and satellite data
- JanVayu cleans and cross-checks them into one picture you can trust

Scattered official data, turned into something a citizen can actually read.

— CHAPTER 02

The live picture

02

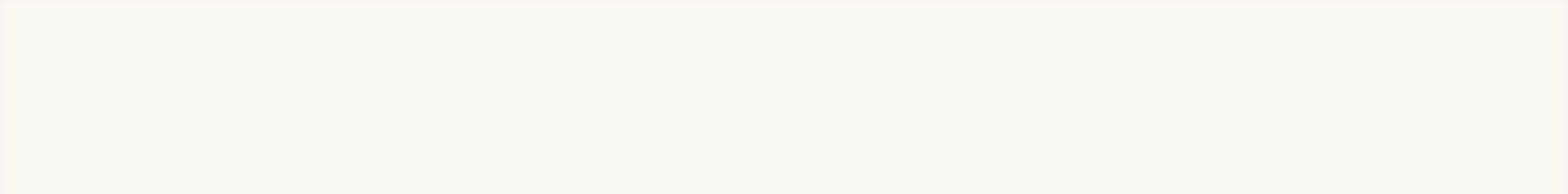
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The dashboard



- Live **PM2.5** for your city in $\mu\text{g}/\text{m}^3$, with the AQI beside it and a plain-language health band
- Cigarette-equivalent and disease-risk read-outs
- “Did you know” sourced facts and an anomaly explainer
- Everything one tap from the city selector at the top

The home screen answers “how bad is it right now, and what does that mean for me?”



— THE LIVE PICTURE

Real-time PM2.5 for 157 cities

- 157 Indian cities, ranked and coloured on **PM2.5** — the fraction that reaches the bloodstream
- Type-to-search — jump to any city instantly
- Covers the major NCAP non-attainment towns
- Every value shown with its source and time

The whole country, one city tap away — and every headline number is PM2.5, not an index.

The live map

- Every monitoring station plotted, colour-coded by AQI
- **MPs / MLAs / Districts** — all 543 Lok Sabha seats and 785 districts coloured by live air; grey where no monitor is near
- **Sources** — landfills, dumpsites, coal mines, red/orange industrial parks, SEZs
- Constituency pop-ups link straight to the Accountability tracker and RTI templates

Where the numbers come from — and the air your MP answers for.

Rankings & comparison

- **City Rankings** — cleanest to worst, live / 7-day / 30-day
- **Compare** — any two cities side by side, with a verdict
- International benchmarks (WHO guideline, world cities)
- Shareable cards for any ranking or comparison

Context turns a number into a story.

Trends & forecast

- **Trends** — scrubbable 24-hour and year-over-year charts
- **Forecast** — a free 5-day PM2.5 outlook (Open-Meteo / CAMS)
- **Farm Fire Tracker** — live stubble fires (NASA FIRMS) across the Punjab-Haryana-NCR belt
- Ask JanVayu can answer “will it be bad tomorrow?”

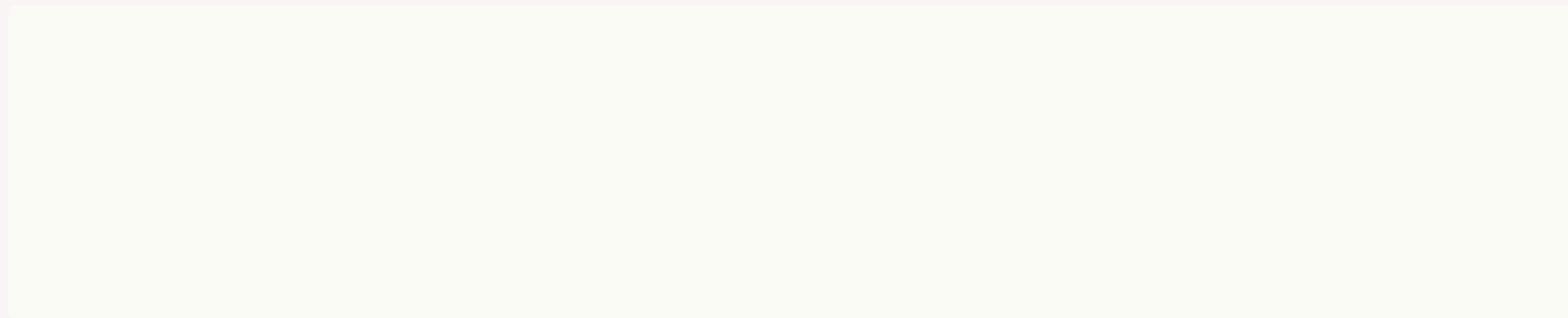
Look back to understand, look forward to plan.

One map, every place in India

- A **Boundaries** menu covering **983,149 areas across seven levels** — state, district, block/mandal/tehsil, gram panchayat and village; city and ward
- A **Colour** menu with nine measures: annual PM2.5, the same air by season, surface heat, tree cover, green cover and built-up
- Every area carries all nine — no dropdown, no list of supported cities
- A tenth, **this year so far**, answers the question everyone asks about the 2024 date — districts only, because it is a ~40 km model corrected against the satellite series, and it rebuilds every month
- **Compare** — plot any two measures against each other for whatever is on screen
- **Who breathes it** — schools and health centres over any level

You should not have to know whether your home is officially a ward or a village to find it.

Every village in India



- All **584,615** village boundaries, one of the seven levels in the same menu
- Each carries the full set — annual and seasonal PM2.5, surface heat, tree, green and built-up cover
- Coloured against the WHO guideline (5) and India's limit (40)
- The live estimate stays separate — and says so when no monitor is near

Two-thirds of India doesn't live in a city. Now it's on the map.



Your airshed, or your town?

- Across **785 districts**, **89.2%** of the variance in annual PM2.5 sits *between* states, not within them
- State medians run Delhi **92.7** down to Ladakh **13.9**; 13 states and UTs are above India's own limit of 40
- Pick a district and the panel splits its distance from the national median into region and local deviation
- **43 years** of PM2.5 underneath it: 783 districts, 1980 to 2022, annual and by season

NCAP sets targets city by city. Most of the burden arrives from the airshed.

Alerts, correlations & the pollution calendar

- **PM2.5 alerts** — Web Push when your city crosses your threshold
- **Pollution Correlations** — how AQI moves with weather and events
- **Pollution Calendar** — the seasonal shape of your city's air
- **Installable app** — add JanVayu to your home screen; alerts arrive even when it's closed

Turn monitoring into something that reaches you.

Health & exposure

03

What the air does to a body

- **Health Impact** — GEMM-based mortality risk for your exposure
- **Beyond the Lungs** — heart, brain, kidneys, metabolism, pregnancy
- Anchored on recent Indian cohort studies, cited in-line
- Translates a $\mu\text{g}/\text{m}^3$ number into a health consequence

PM2.5 is not just a lung problem.

The most vulnerable

- **Children's Health** — smaller lungs, higher dose, developing brains
- **Women's Health** — cooking-smoke exposure, maternal & pregnancy risk
- Household air pollution and the clean-fuel gap
- Each with sourced figures and action steps

Exposure is not shared equally.

Your own exposure

- **Exposure Report** — your estimated daily dose
- **Exposure Diary** — 16 activities weighted by real PM2.5 multipliers
- **Indoor Air** — why inside is often worse, plus a sensor buying guide
- Cigarette-equivalent and life-expectancy read-outs

Make the invisible personal.

What to do about it

- **Should I Go Outside?** — a today-specific yes/no by activity
- **Purifier Calculator** — the right CADR for your room
- **School Closures** — the GRAP thresholds that suspend classes
- **Occupational Exposure** — vendors, traffic police, gig riders, waste workers

From “how bad is it” to “what do I do today”.

The economic bill

The bill nobody sees

- **Economic Cost** — India's ~US\$339.4B / 9.5%-of-GDP loss (Lancet Countdown 2025)
- Lost workdays, lower yields, medical spending
- **Migration & Displacement** — when air drives people to move
- **Migration Calculator** — life-years gained by relocating

Dirty air is a macroeconomic story, not only a health one.

Following the money

05

Policy & budgets

- **Policy Tracker** — NCAP, GRAP, court orders and their status
- **Budget Tracker** — where clean-air money is allocated vs spent
- **City Policy Tracker** — targets and expenditure city by city
- The gap between promise and delivery, made visible

Accountability starts with following the money.

Scorecards & the mission tracker

- **City Scorecards** — a graded report card per city, with one-click RTI
- **Mission Tracker** — independent E1-E5 verification of NCAP claims
- **Political Accountability** — promises vs reality, dated
- **Accountability Brief** — an auto-generated ward-level brief

Turn data into pressure.

Sources, wins & the law

- **Industrial Sources** — the plants and sectors behind the smog
- **Where PM2.5 comes from** — the per-city source-apportionment ring
- **Clean Air Wins** — what has actually worked, honestly
- **Legal Framework** — your rights, key rulings, and how to use them

Name the sources; know your rights.

The RTI Assistant

- Pick a state and a topic; it drafts a filing-ready RTI
- Pre-filled from a scorecard when a city misses its target
- Names the right pollution-control authority
- Turns a citizen's frustration into a formal question

The single most actionable tool on the site.

Citizen action

06

Act, learn, be heard

- **Take Action / Citizen Plan** — concrete steps for your situation
- **Voices & Field Testimony** — the social record and on-ground accounts
- **Workshops** — book a facilitated session, plus a 10-question air-literacy self-check
- **Learning Games** — seven original India-context games

A platform you use, not just read.

News, in context

- **Social Feed** — Reddit, YouTube and Indian news, refreshed automatically
- **Live News** — the latest reporting, aggregated
- **Photo Gallery** — “The air, in pictures”, 32 documentary photographs
- **From the blog** — the newest post and the three behind it, on the dashboard

Keep the issue alive between crises.

Tools & knowledge

07

Ask JanVayu

- A chatbot that answers about your city and ward
- Seasonal, forecast and NCAP context built in
- Answers in ten languages, with sources
- Your fastest way into the data

Natural-language access to everything above.

Learn & look things up

- **Understanding AQI** — what the number actually rolls up
- **Glossary** — the vocabulary, one keystroke away (Ctrl-K)
- **Reading List** — a live Zotero library of ~29 studies
- **FAQ** — 27 sourced answers with a live search

From curious to informed.

Take the data with you

- **Downloads** — briefs, datasets, and citation-ready material
- **Data Archive** — a growing record of readings over time
- **Open Data API** — pull the data straight into your own work, no login
- **Embeddable widgets** — drop live AQI or rankings into any site

Built for researchers, journalists and advocates.

Every number, checked against its source

- Every figure the site states is recomputed from the data on **every change**, and the build fails on drift
- A claim we have **retracted** cannot come back — the build refuses it, in five languages
- A periodic deep audit re-verifies statistics against primary sources; findings are logged publicly
- Two honest methods are never collapsed into one, and unverifiable numbers are flagged rather than guessed

Honesty has to survive the plumbing.

We check our own map against the monitors

- Every PM2.5 figure on the site is **modelled** — satellite, reanalysis or reconstruction — and each says so
- Since v26.6.184 there is also a **measured** layer: CPCB's own hourly record, via the XKDR India Air Quality Database (CC BY 4.0)
- For 2024 the satellite map and the station readings agree at **$r = 0.834$** , the satellite about $2 \mu\text{g}/\text{m}^3$ lower
- That gap is the right direction, not an error — a district average includes the fields around the city

A model you never compare against an instrument is a claim, not a measurement.

Was it policy, or was it the wind?

- A city's air can improve because it emitted less — or because the wind blew harder
- We separate the two for **44 cities, 2018-2024**, using random forests over local weather (Grange et al. 2018)
- **33 are improving** once weather is removed. **11 are not.**
- Steepest falls: Meerut -14.6 , Varanasi -14.2 , Lucknow -14.0 $\mu\text{g}/\text{m}^3$ a year. Rising: Chandigarh $+3.1$, Gwalior $+2.4$, Mumbai $+0.8$

Every “air improved by X%” headline rests on not knowing which of the two happened.

Live vs annual — the honest version

- A single day's AQI swings with the weather
- Annual averages are the fair yardstick for comparison
- Ward maps show a live SPREAD, not a per-street truth
- We say so, everywhere the distinction matters

Being right includes being clear about uncertainty.

— GET INVOLVED

Clean air is not a privilege. It is a right.

The platform is open source and the data is yours — pull it, embed it, file an RTI with it, or add your name to the team at github.com/JanVayu/JanVayu.